

THE DJ CRATE LESSON • AI LITERACY

What's in the Crate?



A 45-minute lesson on how AI learns — and where bias comes from. The hook: an AI is a DJ, and training data is the crate. It can only mix what's in the crate, so a one-genre crate means one-genre answers, no matter who's asking.

AT A GLANCE

TIME	MATERIALS	TECH NEEDED	PREP
45 min	Crate Check worksheet (1 per pair; scissors optional)	None — fully unplugged	Print worksheets. That's it.

OBJECTIVES

- Explain how an AI learns: **input** → **patterns** → **output** — prediction from data, not magic.
- Identify who a dataset serves well, who it leaves out, and connect that gap to real AI bias.
- Discuss data gaps collaboratively: say back a partner's idea and build on it.

STANDARDS ALIGNMENT

CSTA 2-IC-21

ISTE 1.1.d

CCSS ELA SL.6.1

Full standards text in the Teacher Guide.

🎯 WARM-UP — THE ONE-GENRE DJ (7 MIN)

Announce that you are **DJ One-Note**, the school's new music-recommendation AI. Your crate holds one kind of song: loud, fast party tracks. Take 3–4 requests and serve every one with total confidence:

"Something chill to study to?" → "Here's 'TURBO TURBO.' It is not chill. It's what I've got."

"A song for grandma's birthday dinner?" → "Say less. 'VOLUME WAR.' Grandma loves bass drops, probably."

"A slow song for the last dance?" → "'NEON STAMPEDE,' played at... the exact same speed."

Then ask: **is DJ One-Note bad at DJing — or is something else broken?** Fish for it: the DJ is doing its job perfectly. The *crate* is the problem.

Bridge: "Last lesson, the genie granted what you SAID. Today: where its answers even come from. Everything an AI 'knows' comes from a crate of examples — its training data. It can remix the crate, but it can't serve what was never in it. Today you'll BE the DJ."

DIRECT INSTRUCTION — NO MAGIC IN THE MACHINE (8 MIN)

Write the chain on the board: **INPUT** → **PATTERNS** → **OUTPUT**.

INPUT — the crate. Training data: the giant pile of examples an AI studies before you ever meet it — songs for a music AI, mountains of human writing for a chatbot. Nobody types "knowledge" in. It studies examples.

PATTERNS — the training. The AI works through the crate and gets good at ONE thing: guessing what usually goes with what. It doesn't understand any of it — it finds patterns.

OUTPUT — the mix. Ask for something and it predicts what best matches its crate's patterns. A prediction, not a lookup — which is why it can sound confident and still be off.

Say it plainly: "No magic in the machine. An AI doesn't KNOW things — it predicts things, from its crate. So two rules run everything: a GAP in the crate is a gap in the AI, and a LEAN in the crate leans every answer. That lean has a name: bias."

Quick check: "A photo AI's crate is almost all golden retrievers. A husky walks in — what's it going to call it?" (Its best guess from its patterns: golden retriever. Gap in, gap out.)

ACTIVITY — CRATE CHECK (20 MIN)

1. **Setup (3 min):** Pairs — one partner is the **AI DJ**, one runs the **Request Line**. The 12 song cards ARE the DJ's training data. House rules: serve only from the crate, and never refuse — serve the closest thing you've got.
2. **Serve the Request Line (9 min):** Work the six printed requests (Keisha, Nia, Tomás, Priya, Malik, Jordan), swapping roles after three. DJ picks the best card; both rate the fit. End with one live request from a neighboring pair.
3. **Crate Gap Report (5 min):** Pairs record who the crate serves well and who it leaves out. Discussion rule, enforced out loud: *before you add your idea, say back your partner's and build on it*.
4. **Doctor the Crate (3 min):** Invent 3 new cards that fix the biggest gaps. Push past "add one slow song" — which requests still have nothing?

DEBRIEF — THE REAL CRATES (5 MIN)

Pairs call out one gap they found. Connect: "Your crate failed Tomás and Priya. Real crates fail real people the same way." Walk the three examples in the Teacher Guide, then land the big question: "**Nobody put bad songs in your crate — so where did the bias come from?**" (From what got collected and what didn't.)

EXIT TICKET (5 MIN)

Bottom of worksheet: complete the **input** → **patterns** → **output** chain, then diagnose a lopsided homework-helper AI using the word "crate."

DIFFERENTIATION

- **Support:** Sentence frames + pre-sorted crate note (Teacher Guide, p.2).
- **Extension:** Design a 12-card crate that serves every request on the sheet — then defend the trade-offs.
- **ELL:** Ratings are check-marks, not sentences; invented cards may be in any language (which fixes a real gap).
- **No-device classrooms:** Built in — the whole lesson is paper, pairs, and one loud imaginary DJ.